

# A 128-kbit Approximate Search-Capable Content-Addressable Memory (CAM) With Tunable Hamming Distance

Esteban Garzón<sup>1</sup>, Senior Member, IEEE, Eyal Rechef, Roman Golman<sup>2</sup>, Graduate Student Member, IEEE, Odem Harel<sup>1</sup>, Graduate Student Member, IEEE, Yuval Harary<sup>1</sup>, Paz Snapir<sup>1</sup>, Marco Lanuzza<sup>1</sup>, Senior Member, IEEE, Adam Teman<sup>1</sup>, Senior Member, IEEE, and Leonid Yavits<sup>1</sup>, Member, IEEE

**Abstract**—The growing need for approximate matching in data-intensive applications, such as data analytics, machine learning, deep learning, and computational genomics has driven the proposal of our Hamming distance (HD) tolerant content-addressable memory (HD-CAM). HD-CAM features a modified NOR-type associative memory cell that leverages the discharge speed of the matchline (ML) to directly measure the HD between the stored and the query patterns. This novel approach enables efficient in-memory approximate matching. The proposed design was fabricated in a 65-nm technology, running at 125 MHz with an operating voltage of 1.2 V and consuming approximately 0.2 fJ/bit/search at room temperature. HD-CAM features a user-programmable HD tolerance threshold, making it particularly efficient for compare-intensive applications, such as genome analysis, text processing, and database processing. Silicon measurements demonstrate that HD-CAM maintains  $F_1$  score above 90% under process, voltage, and temperature (PVT) variations across a range of HDs.

**Index Terms**—Approximate search, associative memory, content addressable memory (CAM), Hamming distance (HD), similarity search.

## I. INTRODUCTION

ASSOCIATIVE or content-addressable memories (CAMs) are used for a broad range of applications that require compare-intensive operations [1], [2]. In addition to their most pervasive use in network routers, CAMs have been considered in applications, such as genomic analysis, machine learning, reconfigurable computing, and big data workloads [1], [3], [4], [5], [6], [7], [8], [9], [10]. Many of these applications require similarity search, which implies tolerance to a certain

Hamming distance (HD) between the search query pattern and the stored dataset items.

Conventional CAMs typically support only exact matching. However, several approximate CAM designs have been explored to address this limitation. These include error-tolerant CAMs, minimum HD search CAMs, sample time tunable-based CAMs, locality sensitive hashing (LSH), and tunable matchline (ML) discharge rate CAMs [9], [11], [12], [13], [14], [15], [16], [17]. Nevertheless, prior solutions often face issues, such as limited HD tolerance and sensitivity, extremely complex circuitry, or the requirement of preprocessing, making them inefficient for applications that need to tolerate large HDs, such as genome classification and error-correction code (ECC)-enabled fault tolerance.

This work advances the HD tolerant CAM (HD-CAM), originally proposed in [18], from a theoretical concept to a fully functional macro integrated into a system-on-chip (SoC) with a RISC-V processor. The system was fabricated in a 65-nm CMOS process and experimentally validated in silicon. The approximate search operation of the proposed design relies on the fact that the discharge speed of a precharged ML during a *mismatch* is proportional to the HD between the search and stored patterns. HD-CAM is based on a conventional nine-transistor (9T) NOR-type CAM bitcell with an additional device for controlling the ML discharge rate, which facilitates the approximate search capability. The proposed system uses the low-area, low-complexity ML sensing scheme, introduced in [19], to enable efficient, variation-tolerant operation in silicon.

The fabricated 128-kbit macro comprises four  $512 \times 64$  banks, featuring a silicon footprint of 0.21 mm<sup>2</sup> and consuming 0.76 mW per search operation at 1.2 V and room temperature, while operating at a search frequency of 125 MHz. Measurements demonstrate 100% search sensitivity for approximate search scenarios with configurable HDs across different chips, including supply voltage and temperature variations. These features position HD-CAM as a reliable solution for approximate search operations. To our knowledge, this work presents the first silicon-demonstrated, configurable HD-tolerant CAM, overcoming the limitations of existing approaches.

Received 16 June 2024; revised 14 November 2024; accepted 11 January 2025. Date of publication 23 January 2025; date of current version 25 July 2025. This article was approved by Associate Editor Meng-Fan Chang. This work was supported in part by the European Union's Horizon Europe Programme for Research and Innovation under Grant 101047160 (BioPiM) and in part by the Lise Meitner under Grant 1001569396 (AutoPiM). (Corresponding author: Esteban Garzón.)

Esteban Garzón and Marco Lanuzza are with the Department of Computer Engineering, Modeling, Electronics and Systems Engineering (DIMES), University of Calabria, 87036 Rende, Italy (e-mail: esteban.garzon@unical.it).

Eyal Rechef, Roman Golman, Odem Harel, Yuval Harary, Paz Snapir, Adam Teman, and Leonid Yavits are with the Emerging Nanoscaled Integrated Circuits and 774 Systems (EnICS) Laboratory, Faculty of Engineering, Bar-Ilan University, Ramat Gan 5290002, Israel.

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JSSC.2025.3529715>.

Digital Object Identifier 10.1109/JSSC.2025.3529715

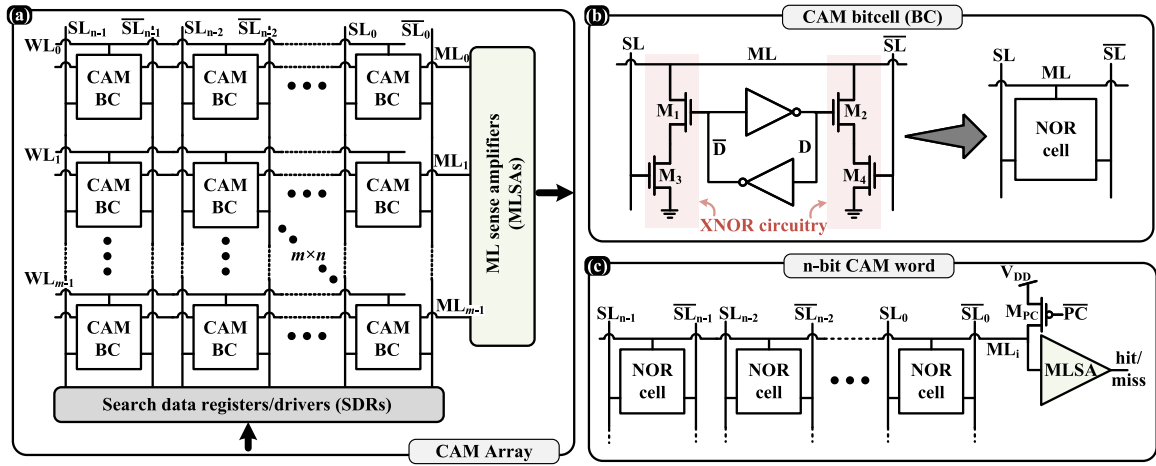


Fig. 1. (a) CAM array along with the search data registers/drivers (SDRs) and MLSAs [18]. (b) NOR bitcell [1]. (c) NOR ML [20]. For the sake of simplicity, WL are not shown in (b) and (c).

The specific contributions of this work include the following:

- 1) A 128-kbit fully functional approximate search CAM with configurable HD tolerance is demonstrated in silicon for the first time.
- 2) Different from state-of-the-art tunable timing-based solutions, HD-CAM exploits the discharge speed of the ML to enable approximate search.
- 3) The fabricated HD-CAM macro achieves 100% sensitivity for different target HD tolerance threshold levels and demonstrates strong discriminative performance, with AUC values exceeding 90%. This capability is maintained under process, voltage, and temperature (PVT) variations by tuning control voltages.

The rest of the paper is organized as follows. Section II presents the background and motivation of our work. Section III discusses the HD tolerant CAM along with its silicon implementation. Section IV shows the silicon measurement evaluation results. Section V concludes this work.

## II. BACKGROUND AND MOTIVATION

### A. Conventional Content-Addressable Memory (CAM)

A CAM is a specialized memory array optimized for fast search operations. Unlike traditional RAM that retrieves data specified by an address, a CAM searches the entire memory array simultaneously to find entries matching a specific query. Fig. 1(a) illustrates the composition of a CAM array, consisting of bitcells arranged in rows and columns [1]. Each CAM bitcell stores a single data bit and can compare it with the corresponding bit in the query data during a search. A key component is the ML, which is shared by all bitcells of a  $n$ -bit word and fed to an ML sense amplifier (MLSA).

Reading from and writing to a CAM bitcell are similar to those of a conventional six-transistor static random access memory (6T-SRAM). During a write operation, the data are driven onto the searchlines (SLs) from the search data register/driver (SDR), and the corresponding word line (WL) is asserted. For reading, data from the accessed row is driven onto the SLs and latched by the search data registers (SDRs).

The associative search operation of all  $n$ -bit CAM words is performed in a single clock cycle. The query data is loaded onto the SLs (SL and  $\overline{SL}$ ), and the MLSA senses the state of the MLs. Depending on whether it is a *match* or *mismatch* between stored and query data, the MLSA in each row outputs a hit or miss.

Many ternary and binary NOR-based and NAND-based CAM cell designs have been proposed in the literature [1], [21], [22], [23], [24], [25], [26]. NAND-type CAMs are energy-efficient, but suffer from high search/compare latency. Conversely, NOR-based CAMs consume more energy but provide high-performance search operations. Fig. 1(b) shows a ten-transistor (10T) NOR CAM bitcell with WLs and 6T-SRAM access transistors omitted for the sake of simplicity. The data nodes  $D$  and  $\overline{D}$  of the 6T-SRAM cross-coupled inverter core are connected to an XNOR circuitry, implemented by the transistors  $M_{1-4}$ . The XNOR operation is carried out between  $D$  and  $\overline{SL}$ , asserting  $M_1/M_3$  or  $M_2/M_4$  transistors to discharge the ML to ground. The ML implements a wired NOR among the bitcells of a CAM row as illustrated in Fig. 1(c). Note that a variation of the 10T NOR CAM cell is the 9T alternative. The latter reduces the bitcell area footprint by using fewer transistors in the ML pull-down paths [refer to XNOR circuitry in Fig. 1(b)], at the expense of longer ML discharge time [1].

The NOR-type CAM carries out the compare operation in two phases: precharge and evaluation. To initiate the precharge phase, the gate of the  $M_{PC}$  transistor is asserted, driving the ML high. During the evaluation phase, an inverted search pattern is driven onto the SLs. If the stored data ( $D$ ) is equal to the inverted search data ( $\overline{SL}$ ) within a single bitcell, the ML is discharged to ground through one of its pull-down paths ( $M_1/M_3$  or  $M_2/M_4$ ). Therefore, at the end of the compare phase, the MLSA signals a *mismatch*. Otherwise, if  $D$  does and  $\overline{SL}$  are unequal for all cells, the ML cannot discharge and remains high, signaling a *match* at the output of the MLSA. This functionality carries out an exact match operation, as provided by conventional CAMs. For the implementation of an approximate CAM, different approaches need to be considered.

### B. Approximate Content-Addressable Memory

Several applications, such as text processing, text retrieval, signal processing, computational biology [1], [3], [27], and emerging genome analysis algorithms [8], [10], require approximate rather than exact search mainly due to errors in the data patterns. In approximate search, a comparison is considered a *match* if the difference between the stored and query patterns is below a predefined threshold. Several approximate search CAM classes follow:

1) *Error Tolerant CAM*: A CAM design that provides soft-error tolerance using ECC through memory redundancy. These designs tolerate a limited HD of 1–4 bits [11], [28]. Additional works include error detection by duplication [12], ECC schemes for multiple cell upsets [29], and error detection and correction using bloom filters [30]. DRAMA [31] and MajorK [32] are unmodified commodity DRAM-based error-tolerant CAM platforms designed for genome classification applications.

2) *Minimum HD Search CAM*: A CAM design that searches for a query pattern with HD below a certain threshold. Such a CAM, utilizing digital circuitry for bit comparison with winner-take-all functionality, is proposed in [33]. HD approximation in memristive crossbar based CAM is proposed in [9] and [34]. Other related works include PPAC [35], and NCAM [36]. PPAC calculates Hamming similarity by performing population count, while NCAM utilizes near-memory logic to calculate data word differences. In RAMANN [6], a CAM for neural network computations, matching and mismatching cells create a voltage divider such that the ML voltage is proportional to the number of matching bits. Liu et al. [14] proposed a multi-state HD CAM with a programmable threshold.

3) *Sample Time Tunable Based CAM*: A CAM design that utilizes the discharge speed of the ML or the score signal delay as a measure of HD. In [37], a delay is added every time a bit mismatch occurs. Therefore, the overall delay of the score signal is proportional to the HD between the search pattern and stored data. In [38], an approximate search-capable CAM with a limited HD tolerance (of up to 2 bits) is proposed to improve the energy efficiency of GPUs. This approach requires meticulous timing of the ML discharge, achieved by dedicated sensing circuitry and precise circuit sizing. Such techniques are subject to false matches and mismatches, limiting their sensitivity and precision [38]. As a result, the approximate search is only partially feasible. Multiple-access, single-charge circuitry, along with a refresh mechanism, is proposed in [39], allowing search operations with an HD tolerance of 2. In [15], different supply voltage levels generate sampling times that determine HD tolerance (up to 8) of a resistive-based ternary CAM (TCAM). Another sampling-time-based approximate TCAM tolerating HD of 6 is presented in [40].

4) *Locality Sensitive Hashing*: CAM uses LSH to implement similarity search [16], [41]. Such an approach can tolerate large HDs; however, it requires the hashing of data before storage and search. Moreover, LSH features a significant false positive rate (FPR) [42], limiting the precision of LSH-based approximate search solutions.

5) *Tunable ML Discharge Rate CAM*: Recent studies have exploited the CAM ML discharge rate to perform approximate matching operations [13], [17], [18], [43]. In [18], an HD tolerant CAM is proposed, where the HD tolerance threshold mainly depends on the sense amplifier (SA), speed of the ML discharge, and the reference voltage of the SA. Further exploring this approach, Merlin et al. [13] and Hanhan et al. [43] proposed edit distance-tolerant genome classification solutions based on RRAM and SRAM, respectively. In [44], the HD-CAM concept is used to allow configurable minimum HD coding, thus improving error tolerance in CAM.

### III. PROPOSED HD TOLERANT CAM

This section describes the circuit design and operation of the proposed HD-CAM solution for approximate search. The HD-CAM macro was custom-designed in a 65-nm CMOS technology, integrated within a system-on-chip (SoC) platform, and fabricated as part of a research test chip [45]. The fabricated HD-CAM macro consists of four 32-kbit banks, organized as 512 64-bit rows (words). Fig. 2(a) shows the architecture of a single HD-CAM bank, including the write and search peripherals<sup>1</sup> and the precharge circuitry. Fig. 2(b) shows the layout of a single HD-CAM bank in a 65-nm CMOS process. The HD-CAM bank design encompasses all necessary peripheral circuitry and occupies a total area of 0.21 mm<sup>2</sup>.

#### A. HD-CAM Bitcell

The 10T HD-CAM bitcell is based on the conventional 9T NOR CAM [1], and is detailed in Fig. 3(a) [18]. The data storage nodes of the SRAM core,  $D$  and  $\overline{D}$ , drive the gates of a pair of nMOS transistors,  $M_1$  and  $M_2$ , which are connected to the SLs on one side, and share a common node on the other side. This shared node is high (“1”) if either  $D$  and  $\overline{SL}$  are “1,” or  $\overline{D}$  and  $SL$  are “1.” In other words, it implements XNOR logic between the stored bit and the inverted query bit at the gate of  $M_3$ , which is connected between ML and  $M_{eval}$ . Therefore,  $M_3$  opens an ML discharge path if a *mismatch* occurs between the query bit (SL) and the stored bit ( $D$ ).

The key addition of the proposed cell is the  $M_{eval}$  transistor, as detailed in [18]. By controlling the adjustable voltage ( $V_{eval}$ ), HD-CAM can be configured to perform an approximate search with different (programmable) HD tolerance thresholds. Lowering  $V_{eval}$  slows down the ML discharge, enabling HD-CAM to match query data whose HD from the stored data is within such HD tolerance threshold. The proposed HD-CAM bitcell exhibits a compact footprint of just 3.24  $\mu\text{m}^2$ , as highlighted in the inset of Fig. 2(b). Differently, the use of the 10T NOR scheme [see Fig. 1(b)] would require two  $M_{eval}$  transistors, leading to a 24% increase in area footprint as compared to the 10T HD-CAM cell.

#### B. HD-CAM ML Sensing Circuitry

Each HD-CAM bank employs an ML sensing scheme based on a positive feedback SA controlled by a replica

<sup>1</sup>The read peripherals like bitline SA and bitline control are not shown for simplicity.



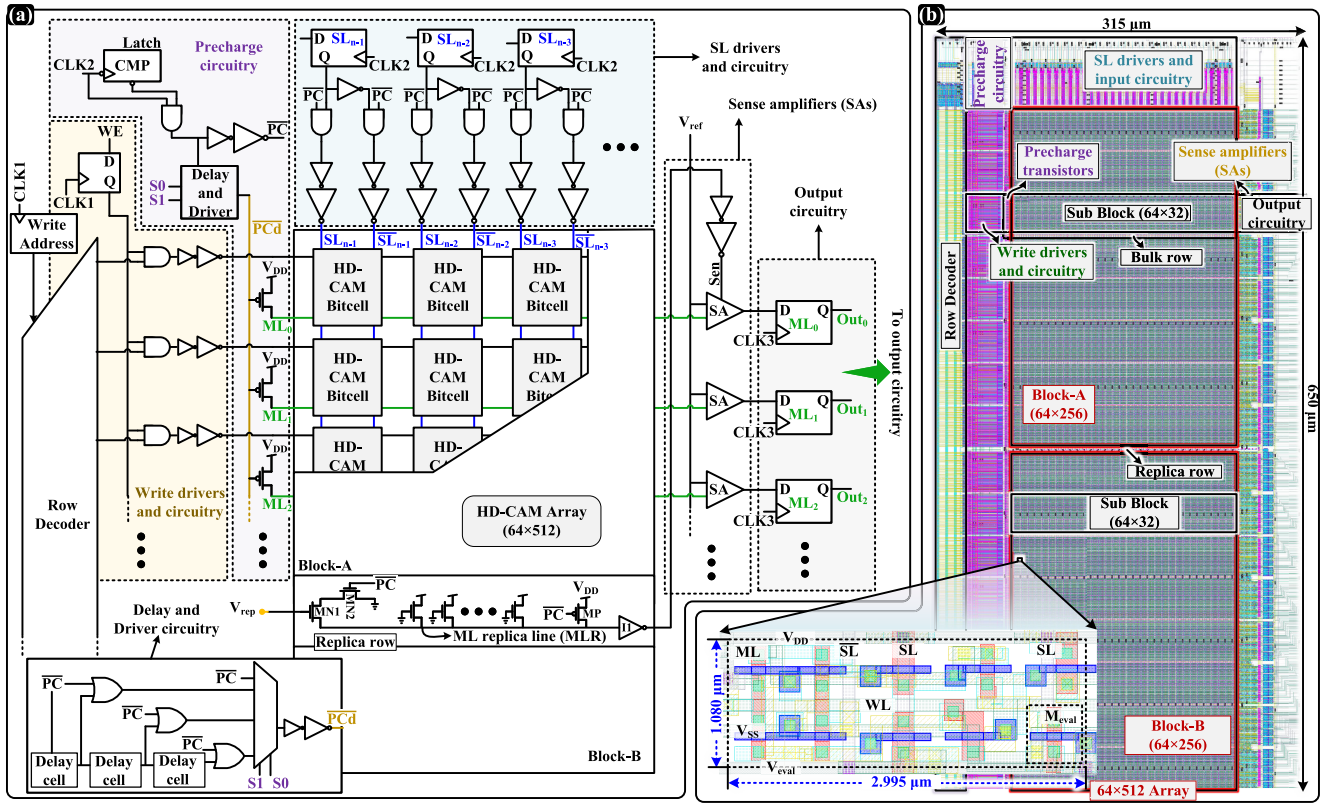


Fig. 2. (a) High-level architecture and (b) layout of a  $62 \times 512$  HD-CAM bank. The 32-kbit HD-CAM bank is organized as two blocks along with peripherals. In the inset of (a) and (b), the 10T HD-CAM bitcell and bitcell layout are shown, respectively.

row [19]. This approach significantly reduces the complexity of the original sensing scheme of [18], by replicating the ML capacitance of a single HD-CAM row and providing PVT-tolerance. A brief description of the ML sensing circuitry follows.

Fig. 2(a) shows the replica row, which is mainly composed of 64 parallel nMOS transistors connected in parallel, one per column, all connected to ground at the gate and  $V_{DD}$  at the drain, while sharing the ML replica (MLR) line at their sources. During the precharge phase ( $\overline{PC} = 0$ ), the MLR charges to  $V_{DD}$ , while MN2 transistor is disabled to prevent any possible discharge to the ground. During the evaluation phase ( $\overline{PC} = 1$ ), the output of inverter I1 enables the sensing circuitry.

Fig. 3(b) shows the scheme of the positive feedback SA comprising cross-coupled inverters and four enable transistors controlled by Sen signal. Two enable transistors link  $V_{ref}$  and ML to the inverter outputs (MLSo). The total area of the sensing circuitry, including all 512 SAs and replica circuitry, is about  $0.0048 \text{ mm}^2$ . For more detailed insights into the ML sensing scheme, we refer the reader to [19].

### C. Write and Search Peripherals

The write peripherals, as shown in Fig. 2(a), consist of the row decoder, write drivers, and bitline drivers (acting as SL drivers during write operation). The 9-to-512 row decoder receives the write address (WADDR) controlled by clock-1 signal (CLK1). The outputs of the row decoder activate specific rows in the HD-CAM array. The write drivers, enabled

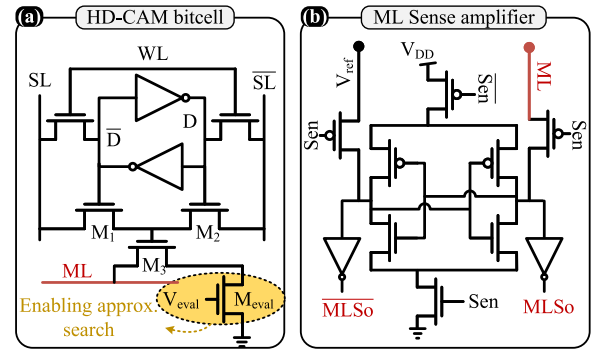


Fig. 3. (a) Structure of the 10T HD-CAM bitcell [18]. Highlighted: nMOS transistor that enables the approximate search through the control of the ML discharge speed. (b) ML sensing amplifier [19].

by a write enable (WE) signal synchronized with CLK1, load the input data,  $D$ . The data to be written onto the HD-CAM array is loaded into the flip-flops within the block of the  $SL$  drivers and circuitry. The  $D$  is controlled by a clock-2 signal (CLK2) and the precharge signal,  $\overline{PC}$ . During write operation,  $\overline{PC}$  is always driven to “1.”

The search-related peripherals comprise the precharge circuitry, SL drivers and circuitry, SAs, and output circuitry. The precharge circuitry consists of a CMP latch that outputs the  $\overline{PC}$  signal followed by drivers of the precharge line that enable the search operation through the two-input AND of the block of the  $SL$  drivers and circuitry. The  $\overline{PC}$  signal also controls the HD-CAM replica row. The  $\overline{PC}$  signal is delayed using the

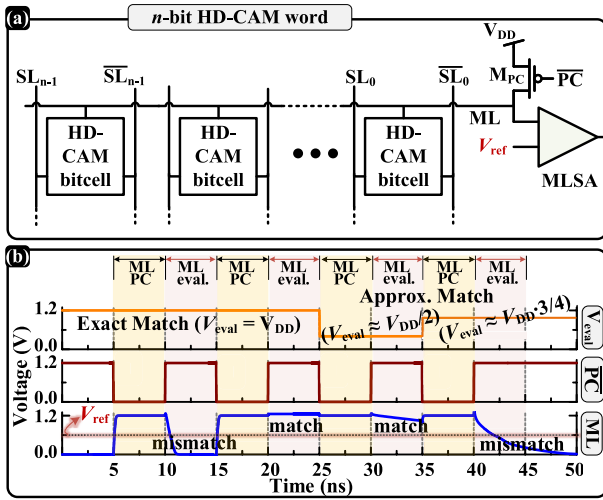


Fig. 4. (a) HD-CAM organized as an  $n$ -bit word. The voltage level of the ML is compared against an MLSA. (b) ML behavior in the case of approximate match and mismatch. Highlighted in (b): the reference voltage ( $V_{ref}$ ) of the MLSA.

*Delay and Driver* circuitry, which contains three delay elements (one per  $\overline{PC}$  delay line), a 4-to-1 MUX, and output drivers. The MUX outputs the  $\overline{PC}$  signal or delayed  $\overline{PC}$  signal ( $\overline{PCd}$ ), depending on the ML precharge delay control signals,  $S_0$  and  $S_1$ . The  $\overline{PCd}$  is connected to the ML precharge transistors of each HD-CAM word. The ML state of the HD-CAM words is compared with a reference voltage ( $V_{ref}$ ) by SAs, one per HD-CAM word. Enabled by the Sen signal, the SAs are responsible for detecting the *match* or *mismatch* between the query pattern and stored information.  $V_{ref}$  has to be properly set by silicon-aware tuning. The outputs of the state of the ML are sampled by clocked flip-flops (CLK3), and then go toward the output circuitry of HD-CAM macro.

#### D. HD-CAM Operation

To illustrate the approximate search operation, consider the HD-CAM organized as an  $n$ -bit word shown in Fig. 4(a). The timing diagrams of Fig. 4(b) show a series of search operations on a single bit storing a “1.” Similar to conventional NOR CAM, HD-CAM carries out a two-phase compare operation controlled by the PC signal: precharge ( $PC = “1”$ ) and evaluation ( $PC = “0”$ ). The first two cycles (5–25 ns) showcase an *exact search operation*, with  $V_{eval}$  set to  $V_{DD}$ . Here, a *mismatch* is detected in the first cycle (5–15 ns) due to the query pattern “0” ( $\overline{SL} = “1”$ ) and the stored value “1.” The XNOR node is driven high and ML is discharged, thereby signaling a *mismatch*. During the second cycle (15–25 ns), the query pattern is “1,” causing the XNOR node to be driven low, thereby cutting off  $M_3$ . Assuming all bits in the row match the query pattern, ML stays high, signaling a *match*.

*Approximate search* is achieved by lowering  $V_{eval}$ , as demonstrated during the next two cycles (25–45 ns). In the third cycle (25–35 ns)  $V_{eval} \approx V_{DD}/2$ . The reduced overdrive of  $M_{eval}$  limits the current while slowing down the discharge rate, thereby maintaining ML above the MLSA sensing threshold ( $V_{ref}$ ) despite the mismatch in the search result. If additional bits mismatch, ML will gradually discharge further, eventually

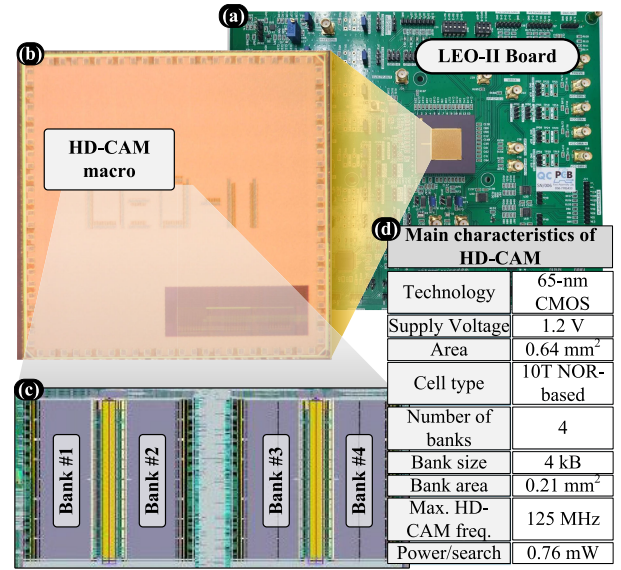


Fig. 5. (a) Fabricated Leo-II SoC board. (b) Test chip micrograph. (c) Layout of full 128-kbit HD-CAM macro. (d) HD-CAM main features.

dropping below  $V_{ref}$ . Raising  $V_{eval}$  to  $\sim 3/4 V_{DD}$  in the fourth cycle (35–45 ns) results in a faster discharge of ML, such that fewer mismatching bits are needed to drive ML below  $V_{ref}$ . Note that, the HD-CAM cell consumes less energy per operation than the conventional 9T and 10T CAM cells [1], as the  $M_{eval}$  transistor limits the ML discharge current during approximate searches, unlike complete ML discharge in the 9T and 10T CAM cells.

Another method to adjust the HD tolerance threshold involves modifying  $V_{ref}$  (the MLSA reference voltage). Therefore, the number of mismatching bits in approximate search (i.e., the HD tolerance threshold) can be configured by tuning the values of  $V_{eval}$  and  $V_{ref}$ .

Note that, because SLs and BLs are shared [see Fig. 3(a)], MLs are discharged if they were previously high and  $V_{eval}$  is high before a write operation. While write operations are less frequent in HD-CAM (similar to most CAM applications such as classification [46], [47]),  $V_{eval}$  is set to “0” during write operations to prevent unnecessary ML discharge.

#### IV. MEASUREMENT RESULTS

HD-CAM was integrated into a 65-nm test chip with additional research blocks. The fabricated SoC features an embedded RISC-V processor [48] connected to peripherals and research blocks through an on-chip AMBA interconnect. HD-CAM is tightly coupled to a local memory farm (XBOX), which provides high-bandwidth memory access to data and registers that are controlled by the host processor or through external peripherals enabling flexible testing of the research blocks. The SoC platform supports a GCC-based software stack for high-level programmability and testing of the fabricated components. A more elaborate description of the test chip architecture can be found in [45]. Fig. 5 presents the measurement setup, SoC board, chip micrograph, full layout of the HD-CAM macro, and the main features of the implemented silicon.

To evaluate the HD tolerance of HD-CAM, random data was generated and loaded into the memory to represent the *stored* dataset. The same data was used as the *query* dataset, with errors injected randomly and uniformly in each 64-bit query word to emulate HD between the query and the stored dataset. During the search process, the *query* dataset was queried in the HD-CAM and the number of matches was counted. This procedure was repeated while varying the HD by incrementally increasing the number of errors from 1 to 20 per 64-bit word. The number of matches (representing the sensitivity of HD-CAM) is then plotted against the HD. All measurements presented below were obtained using the same methodology, showing the measured impact of design variables ( $V_{\text{eval}}$  and  $V_{\text{ref}}$ ) and of PVT variations on the sensitivity of HD-CAM, followed by an elaborate comparison with other approximate search-capable CAMs.

#### A. Design Variables Variations

In this experiment, we expect HD-CAM to tolerate the HD between the dataset and the query and hence we consider every *match* a true positive result, while every accidental *mismatch* is a false negative result. Consequently, the number of matches represents the true positive rate or the *sensitivity* of HD-CAM search, which is plotted in Fig. 6(a) as a function of the sensing threshold ( $V_{\text{ref}}$ ) and HD between the stored dataset and the query. For these measurements, a nominal  $V_{\text{DD}}$  of 1.2 V was used with the evaluation voltage ( $V_{\text{eval}}$ ) set to 0.6 V. The color bar indicates the separation between areas of 100% sensitivity (blue) and 0% sensitivity (yellow), clearly showing stable operation points of 100% sensitivity for large HDs between the query and the dataset. By merely modifying  $V_{\text{ref}}$ , the HD-CAM sensitivity can be configured over a wide range of HDs. A lower  $V_{\text{ref}}$  results in a longer ML discharge to  $V_{\text{ref}}$  level, consequently allowing for the tolerance of higher HDs. The HD tolerance range can also be adjusted by tuning  $V_{\text{eval}}$  at a fixed  $V_{\text{ref}}$ , as shown in Fig. 6(b). By lowering  $V_{\text{eval}}$ , a larger number of mismatching bits are required to discharge ML enough to signal a *mismatch* rather than an approximate match [refer to Fig. 4(b)], thereby enhancing the sensitivity. We also evaluated HD-CAM in terms of precision and  $F_1$  score, as shown in Fig. 6(c). The optimum  $F_1$  score is at  $V_{\text{ref}} = 0.8$  V. These voltage tuples, i.e.,  $V_{\text{eval}}$  of 0.8 V (0.6 V) and  $V_{\text{ref}}$  of 0.8 V, are used in Section IV-B to set the HD tolerance threshold.

To summarize, the sensitivity of a target HD can be tuned by adjusting the reference voltage of the sensing circuitry  $V_{\text{ref}}$ , or by controlling the ML discharge pace by tuning  $V_{\text{eval}}$ . For higher (lower)  $V_{\text{ref}}$  or  $V_{\text{eval}}$ , the HD-CAM sensitivity decreases (increases). Note that, while  $V_{\text{ref}}$  can be used to control the HD tolerance threshold, its ability to reliably achieve large HD tolerance thresholds is limited without slowing the ML discharge rate by lowering  $V_{\text{eval}}$ . By fixing  $V_{\text{ref}}$  and adjusting  $V_{\text{eval}}$ , a broader range of HD tolerance thresholds can be obtained, making  $V_{\text{eval}}$  the more effective and flexible configuration tool.

We extended our experiment to evaluate the impact of true positive rate (TPR), i.e., sensitivity and FPR, i.e., specificity, on the receiver operating characteristic (ROC) and area under the curve (AUC) performance metrics. Fig. 6(d) shows the

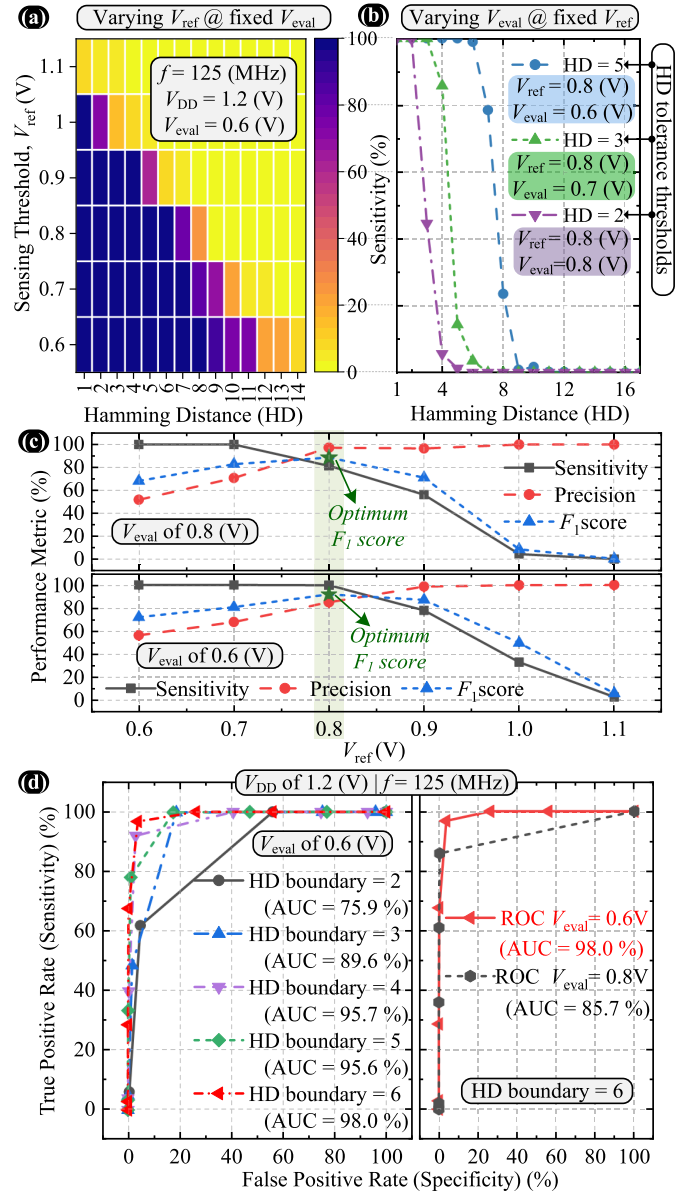


Fig. 6. HD-CAM silicon measurement results: sensitivity for different (a) sensing threshold voltages,  $V_{\text{ref}}$  and (b) evaluation voltages,  $V_{\text{eval}}$ , as a function of the HD. (c) Sensitivity, precision, and  $F_1$  score for different  $V_{\text{ref}}$  voltages. (d) HD-CAM performance results in terms of ROC curves and AUC. The supply voltage ( $V_{\text{DD}}$ ) and operating frequency are 1.2 V and 125 MHz, respectively.

ROC results for a  $V_{\text{eval}}$  of 0.6 V under various tolerance threshold settings, labeled as “HD boundaries,” which establish the decision thresholds distinguishing true positives from FPRs. Fig. 6(d) also includes the AUC of each HD boundary, capturing the overall HD-CAM performance when operating in approximate search mode. The higher the HD boundary, the higher the AUC, reaching up to 98% for the HD boundary of 6, which aligns with the  $V_{\text{eval}}$  setting of 0.6 V. Conversely, a reduced AUC is observed with an increased  $V_{\text{eval}}$  of 0.8 V, as shown in the ROC results on the right in Fig. 6(d).

#### B. Process, Voltage, and Temperature (PVT) Variations

Here we investigate the impact of PVT variations on the HD tolerance (sensitivity, specificity, precision, and  $F_1$  score) of



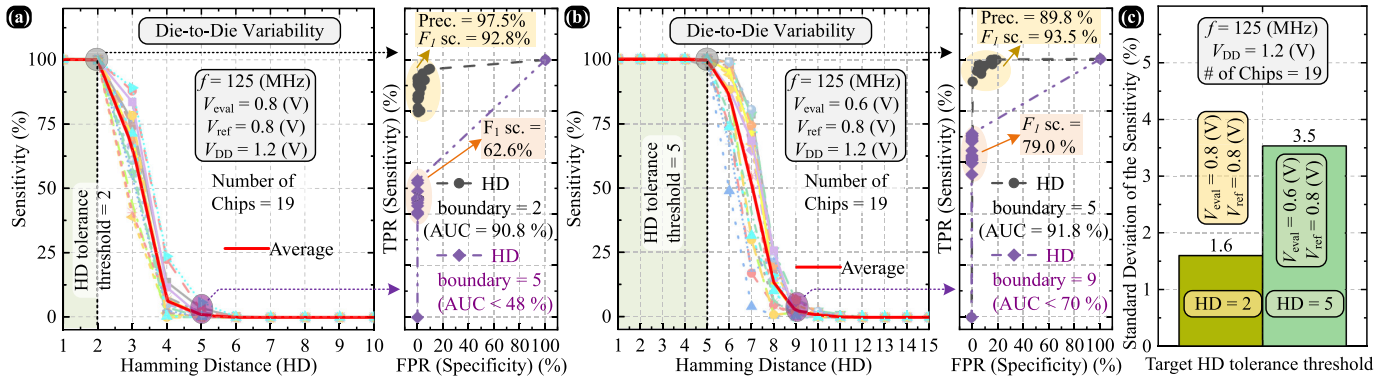


Fig. 7. HD-CAM silicon measurement results: impact of process variability (PV) on the sensitivity and specificity for the target HD tolerance threshold of (a) 2 and (b) 5. ROC, AUC, precision (on average), and  $F_1$  score (on average) metrics shown at the right of (a) and (b). (c) Standard deviation of the HD tolerance thresholds reported in (a) and (b).

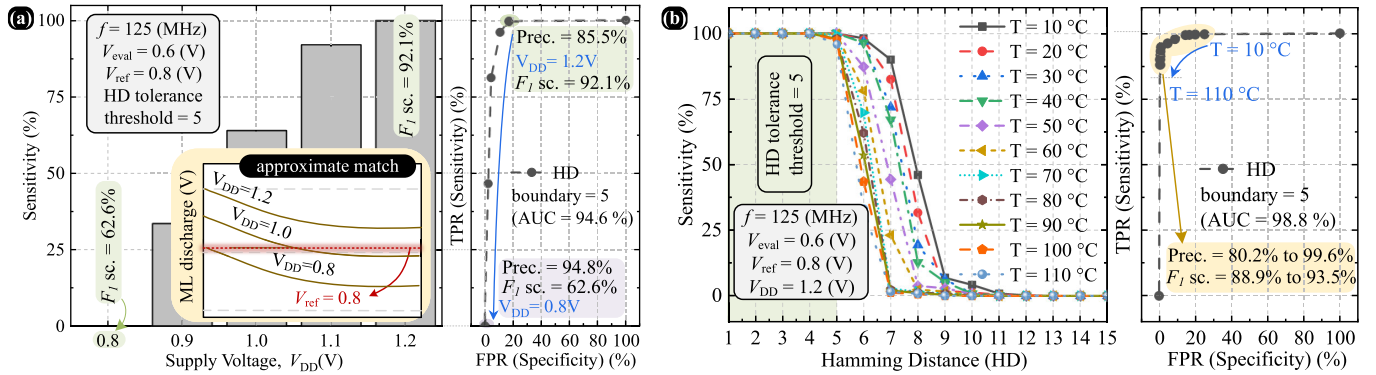


Fig. 8. HD-CAM silicon measurement results for the target HD tolerance threshold of 5. (a) Impact of supply voltage scaling on the sensitivity and specificity. (b) Impact of temperature scaling on the sensitivity and specificity as a function of HD. ROC, AUC, precision, and  $F_1$  score metrics are shown at the right of (a) and (b). The operating frequency,  $V_{\text{eval}}$ , and  $V_{\text{ref}}$  are 125 MHz, 0.6 V, and 0.8 V, respectively.

HD-CAM. Each result was obtained at a single combination of  $V_{\text{ref}}$  and  $V_{\text{eval}}$ , unless otherwise specified.

The effect of die-to-die variability on HD-CAM sensitivity and specificity is presented in Fig. 7(a) and (b) for nineteen distinct chips, operating at 125 MHz, with  $V_{\text{ref}} = 0.8$  V and  $V_{\text{eval}} = 0.8$  V and  $V_{\text{eval}} = 0.6$  V. At this operational setting, all dies achieve an HD tolerance of 2 and 5 with 100% sensitivity, as shown in Fig. 7(a) and (b), respectively. As the HD grows, the sensitivity decreases slightly differently for the measured samples, eventually dropping to 0% at  $\text{HD} \sim 5$  and  $\text{HD} \sim 9$  for all chips in Fig. 7(a) and (b), respectively. In Fig. 7(a) and (b), we evaluated two HD tolerance thresholds, 2 and 5 in Fig. 7(a) and 5 and 9 in Fig. 7(b), in terms of ROC, AUC, precision and  $F_1$  score across all measured chips. The AUC and  $F_1$  score values are high (above 0.9) for HD boundaries of 2 and 5. This result aligns with the 100% sensitivity obtained for the HD tolerance threshold of 2 and 5, respectively. Fig. 7(c) shows the standard deviation of the sensitivity of the HD tolerance threshold reported in Fig. 7(a) and (b). The higher the target HD, the higher the standard deviation of the sensitivity. This is because to obtain a higher HD tolerance, a lower  $V_{\text{eval}}$  voltage is required, moving closer to the threshold voltage of the  $M_{\text{eval}}$  transistor. Therefore, the sensitivity of HD-CAM is reduced.

The supply voltage and temperature variation results are shown in Fig. 8(a) and (b), respectively, for the target HD tolerance threshold of 5 (set by a combination of

$V_{\text{eval}} = 0.6$  V and  $V_{\text{ref}} = 0.8$  V). From Fig. 8(a), the measurements show that sensitivity and  $F_1$  score drop as the supply voltage decreases. This could be expected because ML is precharged to the decreasing  $V_{\text{DD}}$  level while  $V_{\text{ref}}$  remains constant. Hence the discharging ML reaches the  $V_{\text{ref}}$  level quicker (see inset in Fig. 8). From Fig. 8(b), the HD tolerance threshold of 5 is maintained with 100% sensitivity for the full range of measured temperatures. The two ROC curves in Fig. 8(a) and (b) demonstrate high HD-CAM performance, with AUC values of 94.6% and 98.8%, indicating that HD-CAM presents strong discriminative capabilities between matching and mismatching cases under voltage and temperature variations.

Finally, the operation under a range of frequencies is shown with  $V_{\text{eval}} = 0.6$  V and  $V_{\text{ref}} = 0.8$  V in Fig. 9. HD-CAM maintains the HD tolerance threshold of 5 with 100% sensitivity across frequencies from 25 to 125 MHz. The AUC and  $F_1$  score are about 83.8% and 93.6% (on average), respectively.

Note that the maximum HD tolerance threshold that achieves a 100% sensitivity, based on the tested range of  $V_{\text{eval}}$  and  $V_{\text{ref}}$  voltages, is 8. Further tuning of these voltages may enable higher HD tolerance thresholds.

Fig. 10 shows the HD-CAM power breakdown when operating at 125 MHz, 25 °C,  $V_{\text{eval}} = 0.9$  V, and  $V_{\text{ref}} = 0.6$  V with the target HD tolerance threshold of 5. The memory array and

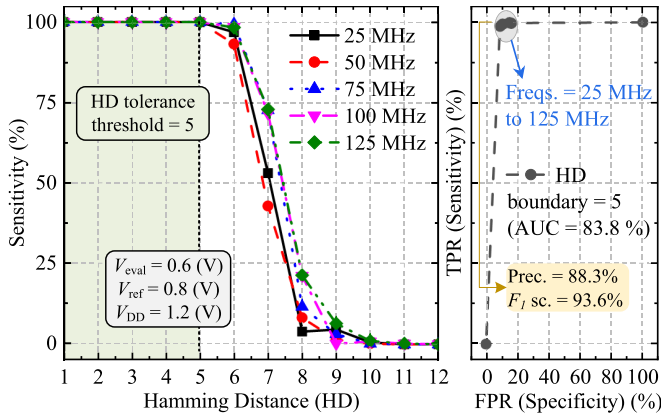


Fig. 9. HD-CAM silicon measurement results: impact of varying frequency on the sensitivity and specificity for the target HD tolerance threshold of 5. ROC, AUC, precision (on average), and  $F_1$  score (on average) metrics are shown at the right.

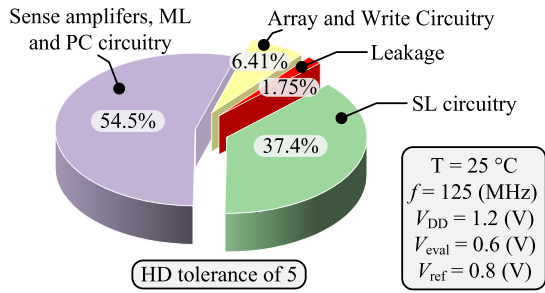


Fig. 10. HD-CAM power breakdown when operating at 25 °C for an HD tolerance threshold of 5. The  $V_{DD}$  and operating frequency are 1.2 V, and 125 MHz, respectively.

write circuitry consume about 6.41%, while the leakage (about 67  $\mu$ W) is 1.75% of the total energy budget which stands at about 0.76 mW. During the precharge (evaluation) phase, SLs are driven to “0” (query data values), consuming about 37.4% of total search power due to their large capacitance and depending on the query pattern. The ML sensing scheme and precharge circuitry account for 54.5%. SL and ML circuitry significantly contribute to the overall power consumption in CAMs [49]. Note that our HD-CAM design objective is a highly accurate and reliable approximate search rather than a low power operation.

Overall, HD-CAM demonstrates robust performance and high reliability under PVT variations. For handling larger voltage and temperature variations, two approaches can be employed to maintain stable HD tolerance thresholds with high sensitivity and AUC: 1) on-chip voltage and temperature monitoring circuitry and 2) dynamic adjustment of  $V_{eval}$  and  $V_{ref}$  through a controller, albeit with larger energy and latency overhead as compared to the first approach, mainly due to the need for real-time sensitivity tracking.

### C. Comparison With Approximate Search-Capable CAMs

Two key observations can be made regarding HD-CAM sensitivity and specificity to process variations, based on both prior findings in [18] and the experimental results presented above. First, while HD-CAM theoretically supports HD tolerance thresholds about half of the word width, as demonstrated in [18], the maximum attainable HD tolerance threshold

TABLE I  
COMPARATIVE RESULTS OF APPROXIMATE SEARCH CAMS

|                                  | [43]   | [39]    | [14]    | This work |
|----------------------------------|--------|---------|---------|-----------|
| <b>Silicon-Proven Technology</b> | NO     | NO      | NO      | YES       |
| <b>Devices per cell</b>          | 65 nm  | 45 nm   | 65 nm   | 65 nm     |
| <b>Supply Voltage (V)</b>        | 42T    | 4T4R    | 2FeFET  | 10T       |
| <b>CAM Type</b>                  | 1.2    | 1.0     | 1.0     | 1.2       |
| <b>HD Tolerance</b>              | Binary | Ternary | Ternary | Binary    |
| <b><math>F_1</math> score*</b>   | 1-12   | 1-5     | 1-8     | 1-8       |
| <b>Word/Rows (bits)</b>          | 73%    | N.S     | N.S     | 91%       |
| <b>Search Time (ns)</b>          | 64/512 | 8/64    | 64/128  | 64/512    |
| <b>Power/search (mW)</b>         | 1.5    | 1.3     | 0.3-0.9 | 8         |
| <b>Energy/bit/search (fJ)</b>    | 30.8   | 154     | 12      | 0.76      |
|                                  | 1.41   | 391     | 0.61    | 0.19      |

\* Average between the HD tolerance. N.S: Not Specified

is limited by the threshold voltage of the  $M_{eval}$  transistor. Lowering  $V_{eval}$  can enable larger HD tolerance. However, as  $V_{eval}$  approaches the threshold voltage of  $M_{eval}$ , the transistors become more susceptible to process variations during approximate match operations leading to an increased number of mismatches (false negatives) detected across HD-CAM words, thereby reducing sensitivity. Low-threshold voltage transistors or wider word widths can theoretically allow larger HD tolerance thresholds. Second, as previously demonstrated in [18] and corroborated by our experimental data, HD-CAM can maintain high sensitivity under process variations by adjusting the voltage tuple  $V_{eval}$  and  $V_{ref}$ . Although the presented measurements are derived from a single batch of silicon dies without skewed corners, high sensitivity and AUC metrics are obtained with proper voltage tuple adjustment in the presence of different process corners.

Table I provides a comparison with relevant state-of-the-art hardware-based approximate CAM alternatives. EDAM [43] leverages conventional CMOS technology to design an approximate search CAM. EDAM exhibits limitations mainly in terms of cross-connectivity among cells, area overhead, and higher power consumption as compared to HD-CAM. Other approximate search CAM designs use RRAM and FeFET technologies [14], [39]. HD-CAM is approximately 2000 $\times$  and 3.2 $\times$  more energy efficient than RRAM- and FeFET-based approximate CAMs, respectively. Approximate CAMs based on RRAM and FeFET have not been silicon-proven, present program/erase cycling degradation, and exhibit reliability issues [50], [51]. HD-CAM represents the first complete silicon-proven system for approximate search CAMs. Under the operating conditions reported in this work (up to 125 MHz,  $V_{ref}$  0.6–0.8 V,  $V_{eval}$  of 0.6–0.9 V, and 10 °C–110 °C), HD-CAM achieves an  $F_1$  score above 90% for approximate search with configurable HD tolerance threshold of up to 8. The proposed design offers a compelling alternative to state-of-the-art solutions by enabling PVT variation-tolerant approximate search functionality using a standard CMOS process. The silicon validation proves the potential of HD-CAM for applications that require, or benefit, from approximate search operations.

### V. CONCLUSION

In this work, we presented a fully operational approximate search CAM macro, fabricated in 65-nm CMOS technology. The proposed HD-CAM enables approximate search



operations while demonstrating robust adaptability to different PVT conditions, and ensuring  $F_1$  score above 90% for a broad range of HD tolerance thresholds. This flexibility is achieved by controlling the ML discharge rate and the sensing threshold of MLSA. HD-CAM extends the conventional NOR-type based CAM bitcell to include an extra transistor that controls the ML discharge rate in the case of a mismatch. The change in the ML is proportional to the HD between the query and stored patterns. When powered at 1.2 V, the 128-kbit HD-CAM operates at 125 MHz, while consuming only 0.76 mW during search at room temperature. Silicon measurements demonstrate the ability of HD-CAM to tolerate different HDs and adjust to process variations and operating conditions. The proposed solution is useful in compare-intensive applications that can benefit from approximate search functionality, such as genome analysis, database processing, and CAM with error correction capabilities. To our knowledge, this work presents the first approximate search-capable CAM demonstrated in silicon.

## REFERENCES

- [1] K. Pagiamtzis and A. Sheikholeslami, "Content-addressable memory (CAM) circuits and architectures: A tutorial and survey," *IEEE J. Solid-State Circuits*, vol. 41, no. 3, pp. 712–727, Mar. 2006.
- [2] B.-D. Yang, Y.-K. Lee, S.-W. Sung, J.-J. Min, J.-M. Oh, and H.-J. Kang, "A low power content addressable memory using low swing search lines," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 58, no. 12, pp. 2849–2858, Dec. 2011.
- [3] R. Karam, R. Puri, S. Ghosh, and S. Bhunia, "Emerging trends in design and applications of memory-based computing and content-addressable memories," *Proc. IEEE*, vol. 103, no. 8, pp. 1311–1330, Aug. 2015.
- [4] E. Garzón, M. Lanuzza, A. Teman, and L. Yavits, "AM4: MRAM crossbar based CAM/TCAM/ACAM/AP for in-memory computing," *IEEE J. Emerg. Sel. Topics Circuits Syst.*, vol. 13, no. 1, pp. 408–421, Mar. 2023.
- [5] W. Li et al., "PIM-WEAVER: A high energy-efficient, general-purpose acceleration architecture for string operations in big data processing," *Sustain. Comput., Informat. Syst.*, vol. 21, pp. 129–142, Mar. 2019.
- [6] M. Ali, A. Agrawal, and K. Roy, "RAMANN: In-SRAM differentiable memory computations for memory-augmented neural networks," in *Proc. ACM/IEEE Int. Symp. Low Power Electr. Design*, New York, NY, USA, Apr. 2020, pp. 61–66.
- [7] M. Imani, Y. Kim, A. Rahimi, and T. Rosing, "ACAM: Approximate computing based on adaptive associative memory with online learning," in *Proc. Int. Symp. Low Power Electron. Design*, 2016, pp. 162–167.
- [8] R. Kaplan, L. Yavits, R. Ginosar, and U. Weiser, "A resistive CAM processing-in-storage architecture for DNA sequence alignment," *IEEE Micro*, vol. 37, no. 4, pp. 20–28, Aug. 2017.
- [9] M. M. Taha and C. Teuscher, "Approximate memristive in-memory Hamming distance circuit," *ACM J. Emerg. Technol. Comput. Syst.*, vol. 16, no. 2, pp. 1–14, 2020.
- [10] R. Kaplan, L. Yavits, and R. Ginosar, "BioSEAL: In-memory biological sequence alignment accelerator for large-scale genomic data," in *Proc. 13th ACM Int. Syst. Stor. Conf.*, 2020, pp. 36–48.
- [11] K. Pagiamtzis, N. Azizi, and F. N. Najm, "A soft-error tolerant content-addressable memory (CAM) using an error-correcting-match scheme," in *Proc. IEEE Custom Integr. Circuits Conf.*, Jul. 2006, pp. 301–304.
- [12] J. A. Maestro, P. Reviriego, S. Baeg, S. Wen, and R. Wong, "Soft error tolerant content addressable memories (CAMs) using error detection codes and duplication," *Microprocessors Microsyst.*, vol. 37, no. 8, pp. 1103–1107, Nov. 2013.
- [13] I. Merlin, E. Garzón, A. Fish, and L. Yavits, "DIPER: Detection and identification of pathogens using edit distance-tolerant resistive CAM," *IEEE Trans. Comput.*, vol. 73, no. 10, pp. 2463–2473, Oct. 2024.
- [14] L. Liu et al., "A reconfigurable FeFET content addressable memory for multi-state Hamming distance," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 6, pp. 2356–2369, Jun. 2023.
- [15] M. Imani, A. Rahimi, P. Mercati, and T. S. Rosing, "Multi-stage tunable approximate search in resistive associative memory," *IEEE Trans. Multi-Scale Comput. Syst.*, vol. 4, no. 1, pp. 17–29, Jan. 2018.
- [16] K. Ni et al., "Ferroelectric ternary content-addressable memory for one-shot learning," *Nat. Electron.*, vol. 2, no. 11, pp. 521–529, 2019.
- [17] Z. Jahshan, I. Merlin, E. Garzón, and L. Yavits, "DASH-CAM: Dynamic approximate Search content addressable memory for genome classification," in *Proc. 56th Annu. IEEE/ACM Int. Symp. Microarchitecture*, Oct. 2023, pp. 1453–1465.
- [18] E. Garzón et al., "Hamming distance tolerant content-addressable memory (HD-CAM) for DNA classification," *IEEE Access*, vol. 10, pp. 28080–28093, 2022.
- [19] E. Garzón, R. Golman, M. Lanuzza, A. Teman, and L. Yavits, "A low-complexity sensing scheme for approximate matching content-addressable memory," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 10, pp. 3867–3871, Oct. 2023.
- [20] R. Taco, E. Garzón, R. Hanhan, A. Teman, L. Yavits, and M. Lanuzza, "Designing precharge-free energy-efficient content-addressable memories," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 32, no. 12, pp. 2303–2314, Dec. 2024.
- [21] A.-T. Do, S. Chen, Z.-H. Kong, and K. S. Yeo, "A high speed low power CAM with a parity bit and power-gated ML sensing," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 21, no. 1, pp. 151–156, Jan. 2013.
- [22] S. Mishra, T. V. Mahendra, and A. Dandapat, "A 9-T 833-MHz 1.72-fJ/bit/search quasi-static ternary fully associative cache tag with selective matchline evaluation for wire speed applications," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 63, no. 11, pp. 1910–1920, Nov. 2016.
- [23] E. Garzón, R. Hanhan, M. Lanuzza, A. Teman, and L. Yavits, "FASTA: Revisiting fully associative memories in computer microarchitecture," *IEEE Access*, vol. 12, pp. 13923–13943, 2024.
- [24] I. Arsovski et al., "1.4Gsearch/s 2-Mb/mm<sup>2</sup> TCAM using two-phase-precharge ML sensing and power-grid pre-conditioning to reduce  $Ldi/dt$  power-supply noise by 50%," *IEEE J. Solid-State Circuits*, vol. 53, no. 1, pp. 155–163, Jan. 2018.
- [25] L. Yavits, A. Morad, and R. Ginosar, "Computer architecture with associative processor replacing last-level cache and SIMD accelerator," *IEEE Trans. Comput.*, vol. 64, no. 2, pp. 368–381, Feb. 2015.
- [26] Q. Dong et al., "A 4 + 2T SRAM for searching and in-memory computing with 0.3-V  $V_{DDmin}$ ," *IEEE J. Solid-State Circuits*, vol. 53, no. 4, pp. 1006–1015, Apr. 2018.
- [27] G. Simoni, F. Reali, C. Priami, and L. Marchetti, "Stochastic simulation algorithms for computational systems biology: Exact, approximate, and hybrid methods," *WIREs Syst. Biol. Med.*, vol. 11, no. 6, p. 1459, Nov. 2019.
- [28] S. C. Krishnan, R. Panigrahy, and S. Parthasarathy, "Error-correcting codes for ternary content addressable memories," *IEEE Trans. Comput.*, vol. 58, no. 2, pp. 275–279, Feb. 2009.
- [29] S. M. Abbas, S. Lee, S. Baeg, and S. Park, "An efficient multiple cell upsets tolerant content-addressable memory," *IEEE Trans. Comput.*, vol. 63, no. 8, pp. 2094–2098, Aug. 2014.
- [30] S. Pontarelli and M. Ottavi, "Error detection and correction in content addressable memories by using Bloom filters," *IEEE Trans. Comput.*, vol. 62, no. 6, pp. 1111–1126, Jun. 2013.
- [31] L. Yavits, "DRAMA: Commodity DRAM based content addressable memory," *IEEE Comput. Archit. Lett.*, vol. 23, no. 1, pp. 65–68, Jan. 2024.
- [32] Z. Jahshan and L. Yavits, "MajorK: Majority based kmer matching in commodity DRAM," *IEEE Comput. Archit. Lett.*, vol. 23, no. 1, pp. 83–86, Jan. 2024.
- [33] H. J. Mattausch, T. Gyohten, Y. Soda, and T. Koide, "Compact associative-memory architecture with fully parallel search capability for the minimum Hamming distance," *IEEE J. Solid-State Circuits*, vol. 37, no. 2, pp. 218–227, Feb. 2002.
- [34] X. Zhu, X. Yang, C. Wu, J. Wu, and X. Yi, "Hamming network circuits based on CMOS/memristor hybrid design," *IEICE Electron. Exp.*, vol. 10, no. 12, 2013, Art. no. 20130404.
- [35] C. C. del Mundo, V. T. Lee, L. Ceze, and M. Oskun, "NCAM: Near-data processing for nearest neighbor search," in *Proc. Int. Symp. Memory Syst.*, Oct. 2015, pp. 274–275.
- [36] O. Castañeda, M. Bobbett, A. Gallyas-Sanhueza, and C. Studer, "PPAC: A versatile in-memory accelerator for matrix-vector-product-like operations," in *Proc. IEEE 30th Int. Conf. Appl.-Specific Syst., Architectures Processors (ASAP)*, vol. 2160, Jul. 2019, pp. 149–156.
- [37] T.-T. Bui and T. Shibata, "A low-power associative processor with the R-th nearest-match Hamming-distance search engine employing time-domain techniques," in *Proc. 5th IEEE Int. Symp. Electron. Design, Test Appl.*, Jan. 2010, pp. 54–57.

- [38] A. Rahimi, A. Ghofrani, K.-T. Cheng, L. Benini, and R. K. Gupta, "Approximate associative memristive memory for energy-efficient GPUs," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Mar. 2015, pp. 1497–1502.
- [39] M. Imani, S. Patil, and T. S. Rosing, "MASC: Ultra-low energy multiple-access single-charge TCAM for approximate computing," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Mar. 2016, pp. 373–378.
- [40] M. Imani, D. Peroni, A. Rahimi, and T. S. Rosing, "Resistive CAM acceleration for tunable approximate computing," *IEEE Trans. Emerg. Topics Comput.*, vol. 7, no. 2, pp. 271–280, Apr. 2016.
- [41] M. S. Riaz, M. Samragh, and F. Koushanfar, "CAMsure: Secure content-addressable memory for approximate search," *ACM Trans. Embedded Comput. Syst.*, vol. 16, no. 5s, pp. 1–20, Oct. 2017.
- [42] G. Marçais, D. DeBlasio, P. Pandey, and C. Kingsford, "Locality-sensitive hashing for the edit distance," *Bioinformatics*, vol. 35, no. 14, pp. i127–i135, Jul. 2019.
- [43] R. Hanhan, E. Garzón, Z. Jahshan, A. Teman, M. Lanuzza, and L. Yavits, "EDAM: Edit distance tolerant approximate matching content addressable memory," in *Proc. 49th Annu. Int. Symp. Comput. Archit.*, Jun. 2022, pp. 495–507.
- [44] Y. Harary, P. Snapir, E. Reshef, E. Garzón, and L. Yavits, "OCCAM: An error oblivious CAM," *IEEE Solid-State Circuits Lett.*, vol. 7, pp. 82–85, 2024.
- [45] E. Garzon et al., "A RISC-V-based research platform for rapid design cycle," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Late Breaking News, May 2022, pp. 2614–2615.
- [46] E. Garzón, L. Yavits, G. Finocchio, M. Carpentieri, A. Teman, and M. Lanuzza, "A low-energy DMTJ-based ternary content-addressable memory with reliable sub-nanosecond search operation," *IEEE Access*, vol. 11, pp. 16812–16819, 2023.
- [47] J. Li, R. K. Montoye, M. Ishii, and L. Chang, "1 mb 0.41  $\mu\text{m}^2$  2T-2R cell nonvolatile TCAM with two-bit encoding and clocked self-referenced sensing," *IEEE J. Solid-State Circuits*, vol. 49, no. 4, pp. 896–907, Apr. 2014.
- [48] Y. Kra, Y. Shoshan, Y. Rudin, and A. Teman, "HAMSA-DI: A low-power dual-issue RISC-V core targeting energy-efficient embedded systems," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 1, pp. 223–236, Jan. 2024.
- [49] I. Hayashi et al., "A 250-MHz 18-mb full ternary CAM with low-voltage matchline sensing scheme in 65-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 48, no. 11, pp. 2671–2680, Nov. 2013.
- [50] B. Zeng et al., "Program/Erase cycling degradation mechanism of HfO<sub>2</sub>-based FeFET memory devices," *IEEE Electron Device Lett.*, vol. 40, no. 5, pp. 710–713, May 2019.
- [51] N. Zagni, F. M. Puglisi, P. Pavan, and M. A. Alam, "Reliability of HfO<sub>2</sub>-based ferroelectric FETs: A critical review of current and future challenges," *Proc. IEEE*, vol. 111, no. 2, pp. 158–184, Feb. 2023.



**Esteban Garzón** (Senior Member, IEEE) received the Ph.D. degree (Hons.) in electronics engineering from the University of Calabria (UNICAL), Rende, Italy, in 2022.

He is currently a Research Fellow with the Department of Computer Engineering, Modeling, Electronics, and Systems Engineering, UNICAL. He has authored or co-authored more than 50 scientific papers in international peer-reviewed journals and conferences, and has participated in several IC tapeouts. His research interests include

domain-specific hardware accelerators, spintronics, cryogenic embedded memories, and standard and emerging technologies for logic and memory, and low-power applications.

Dr. Garzón won a highly competitive research fellowship funded by the Italian Ministry for Universities and Research (MUR), under the call "Horizon Europe 2021–2027 Programme" in 2022.



**Eyal Rechef** received the B.Sc. degree in electrical engineering and physics from Bar-Ilan University, Ramat Gan, Israel, in 2022.

He has been working as a Research Assistant with the Emerging Nanoscaled Integrated Circuits and Systems (EnICS) Laboratory since 2021.



**Roman Golman** (Graduate Student Member, IEEE) received the B.Sc. degree in computer engineering and the M.Sc. degree in electrical engineering from Bar-Ilan University, Ramat Gan, Israel, in 2015 and 2018, respectively, where he is currently pursuing the Ph.D. degree with the Emerging Nanoscaled Integrated Circuits and Systems (EnICS) Laboratory, under the supervision of Prof. Adam Teman.

He has been working as an Intern at "Cadence Design Systems," Petach Tikva, Israel, since 2016.

His research interests include embedded DRAM design and optimization for low-power and high-performance operation, and development of refresh mechanisms for dynamic memories.



**Odem Harel** (Graduate Student Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from Bar-Ilan University, Ramat Gan, Israel, in 2018 and 2020, respectively, where she is currently pursuing the Ph.D. degree.

Her current research interests include design of low-power and energy-efficient embedded memories digital circuits, under varying temperatures and for multiple applications including design flow from circuit-level simulations to layout and chip measurements.



**Yuval Harary** is currently pursuing the B.Sc. degree in computer engineering with Bar-Ilan University, Ramat Gan, Israel, in 2021, with a specialization in hardware and chip design.

Since 2022, he has been actively involved as a Research Assistant with the Emerging Nanoscaled Integrated Circuits and Systems (EnICS) Laboratory, a leading research group at Bar-Ilan University, dedicated to advancing the field of nanoscaled electronics and systems design.



**Paz Snapir** is currently pursuing the Bachelor of Science degree in computer engineering with Bar-Ilan University (BIU), Ramat Gan, Israel, with a focus on hardware and chip design.

Beginning in 2022, he assumed the position of a Research Assistant with the Scale Integrated Circuits and Systems Laboratory (EnICS), BIU.



**Marco Lanuzza** (Senior Member, IEEE) received the Ph.D. degree in electronic engineering from the Mediterranean University of Reggio Calabria, Reggio Calabria, Italy, in 2005.

Since 2006, he has been with the University of Calabria, Rende, Italy, where he is currently an Associate Professor. He has authored over 150 publications in international journals and conference proceedings. His recent research interests include the design of ultralow voltage circuits and systems, the development of efficient models and methodologies for leakage- and variability-aware designs, and the design of digital and analog circuits in emerging technologies.

Prof. Lanuzza is an Associate Editor of Integration, the VLSI Journal.



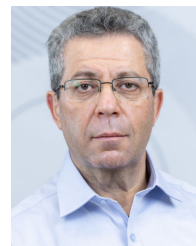
**Adam Teman** (Senior Member, IEEE) received the Ph.D. degree in electrical and computer engineering from Ben-Gurion University (BGU), Be'er Sheva, Israel, in 2014, under a Kreitman Fellowship.

He worked as a Design Engineer at Marvell Semiconductors, Yokneam, Israel, from 2006 to 2007, with an emphasis on physical implementation. From 2014 to 2015, he was a Post-Doctoral Researcher with the Telecommunications Circuits Laboratory (TCL), École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland, under a

Swiss Government Excellence Scholarship. Since 2015, he has been with Bar-Ilan University (BIU), Ramat Gan, Israel, where he is currently an Associate Professor and the Co-Director

of the Emerging Nanoscaled Integrated Circuits and Systems (EnICS) Laboratories. In 2021, he co-founded RAAAM Memory Technologies, Petach Tikva, Israel. He has authored over 125 scientific articles and holds 13 patents. His research interests include embedded memories, energy-efficient circuit design, hardware for artificial intelligence, open-source processor platforms and accelerators, and methodologies for physical implementation.

Dr. Teman has been honored with Teaching Excellence recognitions at both BGU and BIU and has received multiple awards for outstanding research, including the Wolf Foundation Scholarship and the Intel Prize. In 2020, he was awarded the Krill Prize for outstanding young researchers and the BIU Rector's Prize for outstanding research. He is an Associate Editor of the *Microelectronics Journal* and a member of the technical and review boards of several conferences and journals.



**Leonid Yavits** (Member, IEEE) received the M.Sc. and Ph.D. degrees in electrical engineering from Technion, Haifa, Israel.

He is currently with the Faculty of Engineering, Bar-Ilan University, Ramat Gan, Israel. He is a serial entrepreneur who was involved in co-founding and successful management (from a concept to M&A) of several start-ups in the field of application specific integrated circuit (ASIC). His research interests include bioinformatics, domain-specific accelerators, and processing in memory.